CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

CPU & Control Unit

Lecture No.- 01



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Recap of Previous Lecture



Topic Addressing Modes

Topics to be Covered



- **Topic** Addressing Modes
- **Topic** Types of Addressing Modes
- **Topic** CPU Cycle
- **Topic** CPI

Category of addressing mode

Non-Computing
(No any computation needed to calculate effective add.)

- Implied
- Immediate
- Reg.
- Reg. Indirect
- Direct
- Indirect

→ In computation type instns

Computing

- Autoinc. / Autodec.
- Indexed
- PC – Relative
- Base – Reg.

→ support relocation

- branching



Topic : Example

fetch

opcode	mode	add. = 50

mem. ad

	Memory
20	*Opcode* | Mode
21	Address = 50
22	Next Instruction
39	450
40	700
50	80
60	900
72	----------
80	300

PC = ~~20~~ 22

R50 = ~~40~~ 39

Index Reg.

XR = 10

AC

Mode	Effective Address	Operand
1. Immediate Mode	21	50
2. Direct Mode	50	80
3. Indirect Mode	80	300
4. Register Mode	—	40
5. Register Indirect Mode	40	700
6. Autodecrement Mode (Pre)	39	450
7. Indexed Mode	50+10=60	900
8. PC- Relative Mode	22+50=72	—

~~[MCQ]~~

#Q. An instruction is stored at Location 500 with its address field at location 501. The address field has the value 240. A processor register contains the number 160. Evaluate the effective address, if addressing mode is:

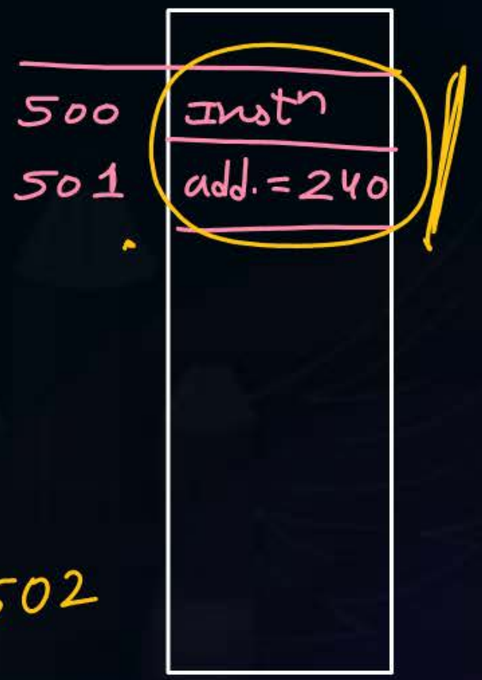
1. Direct ⇒ 240

2. PC-Relative
$= 502 + 240$
$= 742$

3. Immediate ⇒ 501

4. Register Indirect
↳ 160

160
Reg.
PC = 502



[MCQ]

#Q. In case the code is position independent, the most suitable addressing mode is

A Direct mode

B Indirect mode

C Relative mode

D Indexed mode

[MSQ]

#Q. The addressing mode that permits relocation, without any change whatsoever in the code, is

A Indirect addressing

B ✓ Base register addressing

C Indexed addressing

D ✓ PC relative addressing

[MCQ]

#Q. A relative branch mode type instruction is stored in memory at address 390. The branch is made to an address 460.

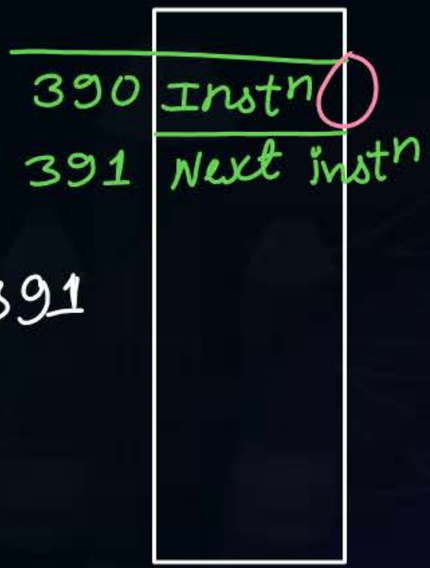
1. What should be the value of relative address field of the instruction?
2. Determine the value of PC before instruction fetch, after the fetch and after execution phase?

E.A. (Target) = 460

Target = PC + offset

460 = 391 + offset

offset = 69



PC = 391

	PC
before fetch	390
after fetch	391
After execution	460

- ~~Fetch~~
- decode
- E.A. Calculatn
- Executn

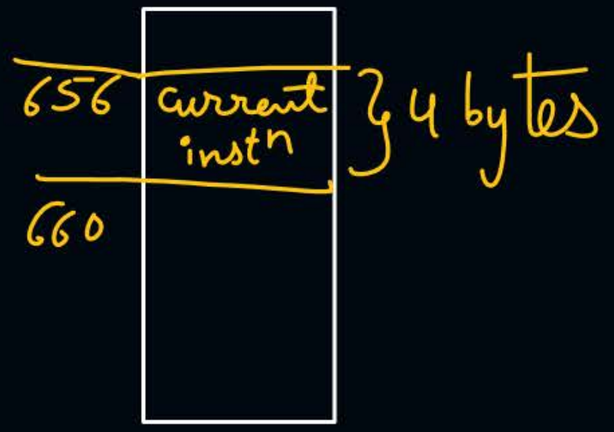
Ques) A PC - relative branch mode type instⁿ takes jump to address 780. The offset is given in instⁿ = 120. If the instⁿ is 4 bytes size and mem. is byte addressable, then starting add. of instⁿ is ___656___ ?

Soln :-

$$780 = PC + 120$$

$$PC = 780 - 120$$

$$= 660$$



[MCQ]

GATE- PYQ

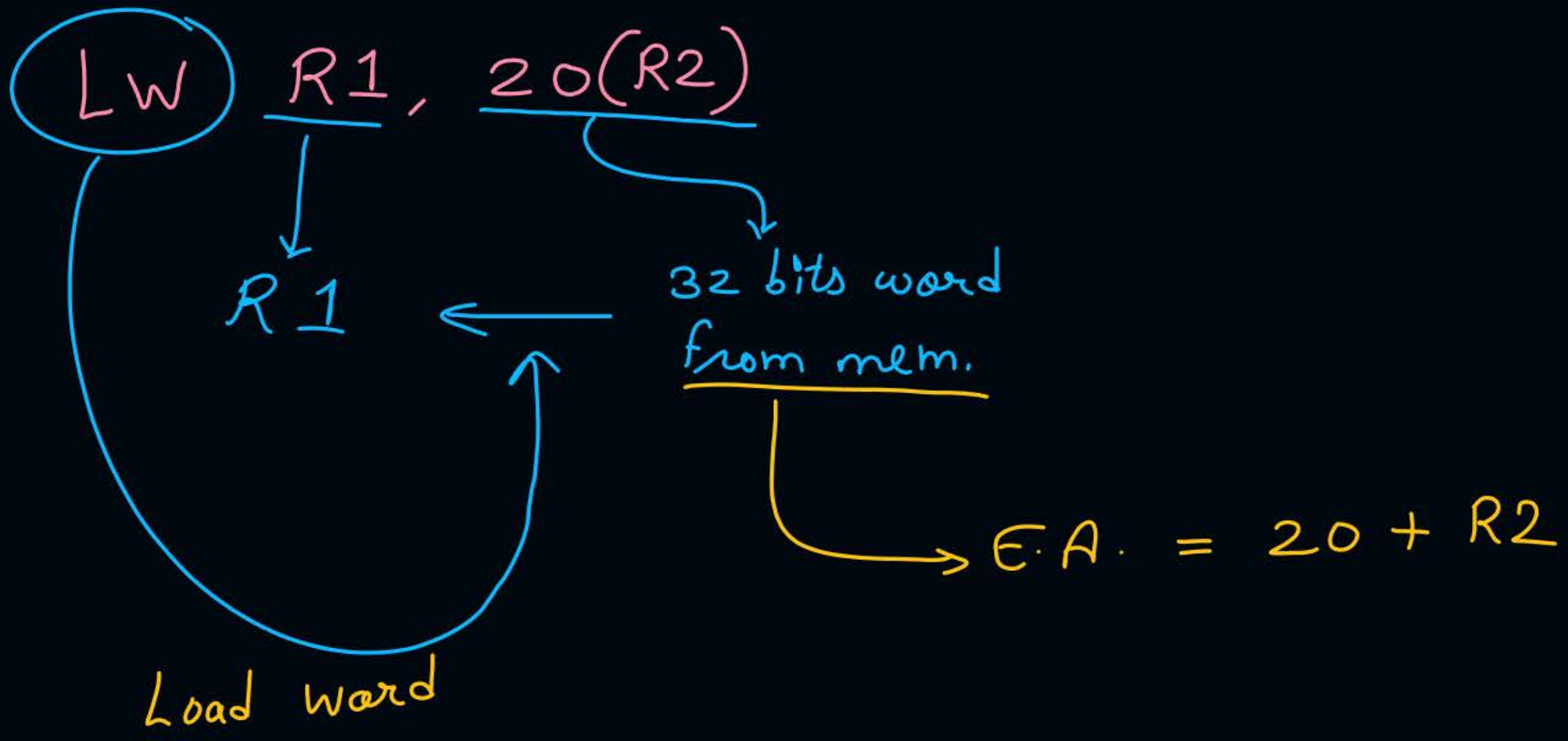
#Q. Consider a hypothetical processor with an instruction of type LW R1, 20(R2); which during execution reads a 32-bit word from memory and stores it in a 32-bit register R1. The effective address of the memory location is obtained by the addition of a constant 20 and the contents of register R2. Which of the following best reflects the addressing mode implemented by this instruction for operand in memory?

A. Immediate Addressing

B. Register Addressing

C. Register indirect addressing

D. Base indexed addressing


LW R1, 20(R2)
R1
32 bits word from mem.
E.A. = 20 + R2
Load word

[NAT]

GATE- PYQ

#Q. Consider a three word machine instruction

Ans = 4

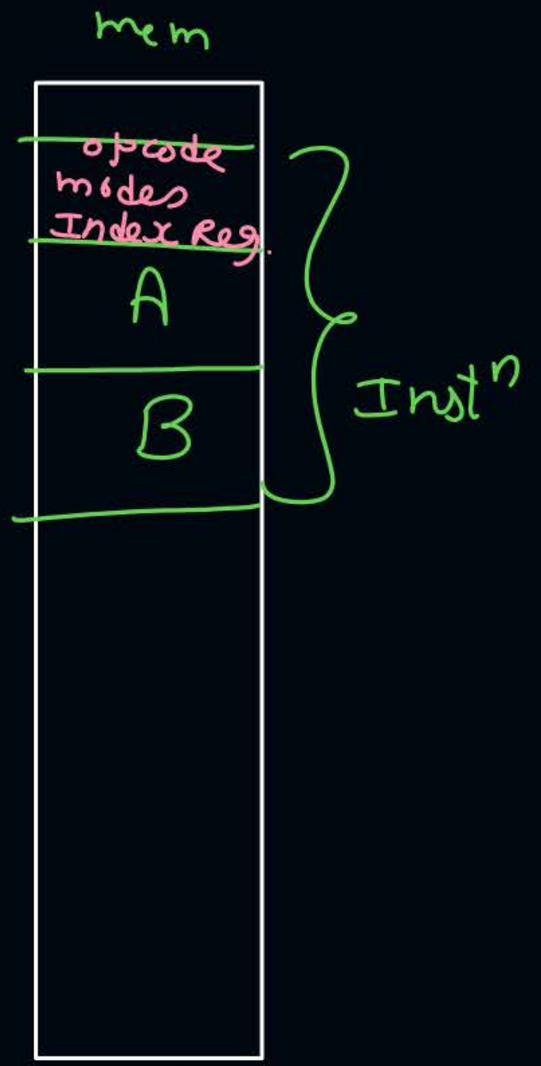
ADD A[R0], @B

The first operand (destination) "A[R0]" uses indexed addressing mode with R0 as the index register. The second operand (Source) "@B" uses indirect addressing mode. A and B are memory addresses residing at the second and the third words, respectively. The first word of the instruction specifies the opcode, the index register designation and the source and destination addressing modes. During execution of ADD instruction, the two operands are added and stored in the destination (first operand).

The number of memory cycles needed during the execution cycle of the instruction is_____.

ADD A[R0], @B





	operand 1	operand 2
1. Decode	–	–
2. E.A. Calculatn & operand fetch	1	2
3. Execution	–	–
4. write back	1	
	4	

[NAT]

H.W.

#Q. Consider a 6-words instruction, which is of the following type:

Opcode	Mode1	Mode2	Address1	Address2

The first operand (destination) uses register indirect mode and second operand uses indirect mode. Assume each operand is of size 2 words, each address is of 2 words and main memory takes 30ns for 1 byte access. Further assume that the opcode denotes addition operation which copies result of addition of 2 operands. One word size is 4 bytes. Total time required in:

memory access

1. Fetch cycle of instruction
2. Execution cycle of instruction
3. Instruction cycle of instruction

Instruction
+
Addressing mode

→ PYQs

$Hz = \frac{1}{second}$

CPU



Topic : CPU

1. CPU Cycle → Time taken by CPU to perform smallest operation.
2. CPU Clock rate (freq.) $= \frac{1}{\text{cycle time}}$ $\Rightarrow$ $1 \text{ cycle time} = \frac{1}{\text{clock rate}}$
3. CPI (Cycles Per Instn) $\Rightarrow$ No. of CPU cycles needed to execute an instruction.
4. Execution Time

(Avg.) 1 instn execution time $= CPI_{avg} * \text{CPU cycle time}$

Program execution time (n instns in program) $= n * CPI_{avg} * \text{CPU cycle time}$

$= \frac{n * CPI_{avg}}{\text{clock rate}}$



Topic : MIPS (million Instns Per Second)

in t time CPU executed no. of instns $= n$

in 1 ——— " ——— $= \frac{n}{t}$

$$= \frac{n * \text{clock rate}}{n * CPI_{avg}}$$

$$= \frac{\text{clock rate}}{CPI_{avg}}$$

$$MIPS = \frac{\text{clock rate}}{CPI_{avg} * 10^6}$$



Topic : Average CPI

Consider computing the overall CPI for a machine A for which the following performance measures were recorded when executing a set of benchmark programs. Assume that the clock rate of the CPU is 200 MHz

Instruction Category	Number of Instructions	No. of cycles per Instruction	Total cycles
ALU	48	1	48*1=48
Load & Store	10	3	10*3=30
Branch	39	4	39*4=156
Other	3	5	3*5=15
	Total=100		249

$$\text{Avg CPI} = \frac{\text{Total cycles to execute all instns}}{\text{total no. of instns}}$$

$$= \frac{249}{100}$$

$$= 2.49$$

$$MIPS = \frac{200\ MHz}{2.49 * 10^6}$$

$$= 80.32\ MIPS$$

[NAT]

GATE- PYQ

#Q. Consider two processors P1 and P2 executing the same instruction set. Assume that under identical conditions, for the same input, a program running on P2 takes 25% less time but incurs 20% more CPI (clock cycles per instruction) as compared to the program running on P1. If the clock frequency of P1 is 1GHz, then the clock frequency of P2 (in GHz) is 1.6 ?

	P1	P2
Time	t_1	$t_2 = 0.75\, t_1$
CPI	C_1	$C_2 = 1.2\, C_1$
n	n_1	n_2
freq.	$f_1 = 1GHz$	$f_2 = ?$

$$n_1 = n_2$$

$$\frac{t_1 * f_1}{C_1} = \frac{t_2 * f_2}{C_2}$$

$$\frac{t_1 * f_1}{C_1} = \frac{0.75\, t_1 * f_2}{1.2\, C_1}$$

$$f_2 = 1GHz * \frac{1.2}{0.75} = 1.6GHz$$

Q.61 An application executes 6.4×10^8 number of instructions in 6.3 seconds. There are four types of instructions, the details of which are given in the table. The duration of a clock cycle in nanoseconds is 3.0 . (rounded off to one decimal place)

GATE – 2025 shift2

Instruction type	Clock cycles required per instruction (CPI)	Number of instructions executed
Branch	2	$2.25 \times 10^8 = 4.5 * 10^8$
Load	5	$1.20 \times 10^8 = 6 * 10^8$
Store	4	$1.65 \times 10^8 = 6.6 * 10^8$
Arithmetic	3	$1.30 \times 10^8 = 3.9 * 10^8$
		Total $= 21 * 10^8$

execution time = n * CPI * cycle time

6.3 seconds $= 6.4 * 10^8 * \frac{21 * 10^8}{6.4 * 10^8} *$ cycle time

Cycle time $= \frac{6.3 \text{ sec}}{21 * 10^8} = \frac{0.3 \text{ sec}}{10^8} = \frac{3 \text{ sec}}{10^9} = 3 \text{ nsec}$



2 mins Summary

Topic Addressing Modes

Topic Types of Addressing Modes



Happy Learning

THANK - YOU